where h is the height above the ground (in meters), g is the acceleration of gravity (9.8 meters per second^2), v is the initial velocity (9.8 meters per second), and t is the time (in seconds).

Now, suppose we want to know *when* the ball passes a certain height. Plugging in the known values and solving for t:

$$t = 1 \pm \sqrt{1 - h/4.9}$$

For instance, the ball is at a height of 3 meters *twice*: $t = 0.38$ (going up) and $t = 1.62$ seconds (going down).

As long as we ask reasonable questions, these equations give reasonable answers. But what happens when we ask unreasonable questions? For example: At what time does the ball reach a height of 10 meters? This question has no answer in reality because the ball *never* reaches this height. Nevertheless, plugging the value of $h = 10$ into the above equation gives two answers: $t = 1 + \sqrt{-1.041}$ and $t = 1 - \sqrt{-1.041}$. Both these answers contain the square-root of a negative number, something that does not exist in the world as we know it. This unusual property of polynomial equations was first used by the Italian mathematician Girolamo Cardano (1501-1576). Two centuries later, the great German mathematician Carl Friedrich Gauss (1777-1855) coined the term **complex numbers**, and paved the way for the modern understanding of the field.

Every complex number is the sum of two components: a **real part** and an **imaginary part**. The real part is a **real number**, one of the ordinary numbers we all learned in childhood. The imaginary part is an **imaginary number**, that is, the *square-root of a negative number*. To keep things standardized, the imaginary part is usually reduced to an ordinary number multiplied by the square-root of negative one. As an example, the complex number: $t = 1 + \sqrt{-1.041}$, is first reduced to: $t = 1 + \sqrt{1.041}\sqrt{-1}$, and then to the final form: $t = 1 + 1.02\sqrt{-1}$. The real part of this complex number is 1, while the imaginary part is $1.02\sqrt{-1}$. This notation allows the abstract term, $\sqrt{-1}$, to be given a special symbol. Mathematicians have long used i to denote $\sqrt{-1}$. In comparison, electrical engineers use the symbol, j, because i is used to represent electrical current. Both symbols are common in DSP. In this book the electrical engineering convention, j, will be used.

For example, all the following are valid complex numbers: $1 + 2j$, $1 - 2j$, $-1 + 2j$, $3.14159 + 2.7183j$, $(4/3) + (19/2)j$, etc. All ordinary numbers, such as: 2, 6.34, and -1.414, can be viewed as a complex number with *zero* for the imaginary part, i.e., $2 + 0j$, $6.34 + 0j$, and $-1.414 + 0j$.

Just as real numbers are described as having positions along a number line, complex numbers are represented by locations in a two-dimensional display called the **complex plane**. As shown in Fig. 30-1, the horizontal axis of the